

##Creation of Tables:

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    manager_id INT,  
    salary INT,  
    hire_date DATE,  
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id),  
    FOREIGN KEY (manager_id) REFERENCES employees(emp_id)  
);
```

```
CREATE TABLE projects (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(50),  
    dept_id INT,  
    budget INT,  
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
);
```

```
CREATE TABLE employee_project (  
    emp_id INT,  
    project_id INT,  
    hours INT,
```

```
PRIMARY KEY (emp_id, project_id),  
FOREIGN KEY (emp_id) REFERENCES employees(emp_id),  
FOREIGN KEY (project_id) REFERENCES projects(project_id)  
);
```

INSERTION OF RECORDS

```
INSERT INTO departments (dept_id, dept_name, location) VALUES  
(10, 'Engineering', 'Chennai'),  
(20, 'Sales', 'Hyderabad'),  
(30, 'HR', 'Bengaluru'),  
(40, 'Research', 'Pune'),  
(50, 'Finance', 'Mumbai');
```

```
INSERT INTO employees (emp_id, emp_name, dept_id, manager_id, salary, hire_date) VALUES  
(101, 'Arjun', 10, NULL, 90000, '2021-01-15'),  
(102, 'Priya', 10, 101, 65000, '2022-03-10'),  
(103, 'Karthik', 20, 106, 55000, '2022-07-20'),  
(104, 'Meena', 30, NULL, 60000, '2020-11-05'),  
(105, 'Rahul', NULL, 101, 50000, '2023-02-12'),  
(106, 'Divya', 20, NULL, 85000, '2019-06-25'),  
(107, 'Sanjay', 10, 101, 72000, '2021-09-17'),  
(108, 'Anitha', 40, NULL, 78000, '2020-04-30'),  
(109, 'Vikram', 20, 106, 48000, '2024-01-08');
```

```
INSERT INTO projects (project_id, project_name, dept_id, budget) VALUES  
(201, 'Banking App', 10, 500000),  
(202, 'Sales Portal', 20, 300000),  
(203, 'AI Research', 40, 800000),  
(204, 'Employee System', 30, 250000),
```

```
(205, 'Cloud Migration', 10, 600000),  
(206, 'Audit Automation', 50, 350000);
```

```
INSERT INTO employee_project (emp_id, project_id, hours) VALUES  
(101, 201, 120),  
(102, 201, 100),  
(102, 205, 80),  
(103, 202, 110),  
(106, 202, 70),  
(107, 201, 90),  
(107, 205, 130),  
(108, 203, 150),  
(104, 204, 60);
```

#####SQL QUERIES

```
1. SELECT e.emp_id, e.emp_name, d.dept_name, d.location  
   FROM employees e  
   INNER JOIN departments d ON e.dept_id = d.dept_id;
```

```
2. SELECT e.emp_id, e.emp_name, d.dept_name  
   FROM employees e  
   LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

```
3. SELECT d.dept_id, d.dept_name  
   FROM departments d  
   LEFT JOIN employees e ON d.dept_id = e.dept_id  
   WHERE e.emp_id IS NULL;
```

4. SELECT e.emp_name, p.project_name, d.dept_name, ep.hours
FROM employee_project ep
INNER JOIN employees e ON ep.emp_id = e.emp_id
INNER JOIN projects p ON ep.project_id = p.project_id
INNER JOIN departments d ON e.dept_id = d.dept_id;

5. SELECT p.project_id, p.project_name, e.emp_name
FROM projects p
LEFT JOIN employee_project ep ON p.project_id = ep.project_id
LEFT JOIN employees e ON ep.emp_id = e.emp_id;

6. SELECT e.emp_name AS employee, m.emp_name AS manager
FROM employees e
LEFT JOIN employees m ON e.manager_id = m.emp_id;

7. SELECT e.emp_id, e.emp_name, COUNT(ep.project_id) AS total_projects
FROM employees e
INNER JOIN employee_project ep ON e.emp_id = ep.emp_id
GROUP BY e.emp_id, e.emp_name
HAVING COUNT(ep.project_id) > 1;

8. SELECT d.dept_name,
COUNT(e.emp_id) AS employee_count,
COALESCE(AVG(e.salary), 0) AS avg_salary
FROM departments d
LEFT JOIN employees e ON d.dept_id = e.dept_id
GROUP BY d.dept_name;

```
9. SELECT e.emp_name,  
    d1.dept_name AS employee_dept,  
    p.project_name,  
    d2.dept_name AS project_dept  
FROM employee_project ep  
INNER JOIN employees e  ON ep.emp_id = e.emp_id  
INNER JOIN projects p   ON ep.project_id = p.project_id  
INNER JOIN departments d1 ON e.dept_id = d1.dept_id  
INNER JOIN departments d2 ON p.dept_id = d2.dept_id  
WHERE e.dept_id <> p.dept_id;
```

```
10. SELECT p.project_id, p.project_name, d.dept_name, p.budget  
FROM projects p  
LEFT JOIN employee_project ep ON p.project_id = ep.project_id  
INNER JOIN departments d ON p.dept_id = d.dept_id  
WHERE ep.emp_id IS NULL;
```